

AURA-8 Architecture Diagram & Technical Specification

Economic Times AI Hackathon 2.0 | Problem Statement 8: AI for Industrial Knowledge Intelligence

1. High-Level System Architecture Diagram

The system is structured across 4 decoupled architectural layers:

CLIENT LAYER (Next.js 15 App Router) • Gateway Landing Page (Demo vs Clean Start) • Persistent Floating AI Chatbot Drawer • Interactive 2D HTML5 Canvas Knowledge Graph • Universal Document Ingester Dropzone	API ORCHESTRATION LAYER • POST /api/upload (PDF stream parsing) • POST /api/query (Gemini RAG call) • GET /api/graph (Knowledge Graph nodes) • POST /api/mode/demo (Sample hydrocracker data)
INTELLIGENCE & VECTOR ENGINE • pdf-parse & PyPDF Text Extractor • Recursive Text Chunker (600ch, 100 overlap) • TF-IDF & Cosine Similarity Vector Index • RegEx & NLP Entity Extractor (Tags P-101A, V-301)	EXTERNAL AI LLM SERVICE • Google Gemini API (gemini-flash-latest) • Endpoint: https://generativelanguage.googleapis.com/ • Header: X-goog-api-key • Output: Grounded Q&A; with page citations

DATA EXECUTION FLOW DIRECTIONS:
User Document Upload ■■■■ Next.js /api/upload ■■■■ pdf-parse Extractor ■■■■ Text Chunker & Vector Store ■■■■ Entity Parser ■■■■ Knowledge Graph
User Chatbot Query ■■■■ Next.js /api/query ■■■■ Vector Cosine Match (Top 3) ■■■■ Google Gemini API (gemini-flash-latest) ■■■■ Grounded Response + Page Citations

2. Ingestion & RAG Execution Sequence Diagram

Step #	Sequence Event	Source → Destination	Description & Payload
1	PDF Document Upload	User → Next.js UI	User drags PDF into dropzone or uploads via file selector.
2	Multipart POST Request	Next.js UI → /api/upload	Posts file FormData to same-origin relative API route.
3	Stream Decompression	/api/upload → pdf-parse	Decompresses /FlateDecode PDF binary stream into clean text.
4	Chunking & Vector Index	/api/upload → Vector Store	Splits text into 600-character overlapping chunks & indexes TF-IDF.
5	Entity & Tag Extraction	/api/upload → Entity Parser	Extracts equipment tags (P-101A) and statutory standards (OISD-137).
6	Graph Population	/api/upload → Knowledge Graph	Populates 2D Knowledge Graph nodes & unlocks dashboard infographics!
7	RAG User Query	User → Persistent Chatbot	User asks technical question in persistent AI chatbot drawer.
8	Cosine Match Search	/api/query → Vector Store	Retrieves top 3 relevant text chunks matching query term frequencies.
9	Gemini LLM Call	/api/query → Gemini API	Posts generateContent prompt with X-goog-api-key header.
10	Grounded Answer Render	Gemini API → Chatbot UI	Renders grounded response with clickable page-level text citations.

3. Technology Stack & Component Responsibilities

Subsystem	Technology Stack	Core Responsibility
Frontend UI	Next.js 15, React 19, Tailwind CSS	Gateway landing page, persistent floating chatbot, interactive Canvas 2D graph

RAG Backend	Next.js API Routes & Python FastAPI	Same-origin /api/upload and /api/query endpoints with zero CORS issues.
LLM Engine	Google Gemini (gemini-flash-latest)	Synthesizing grounded responses, page citations, and executive summaries.
PDF Parser	pdf-parse (Node) & PyPDF (Python)	Decompressing PDF streams, extracting equipment tags (P-101A) and OISD c